Def.
Let G be a group, and $A \subseteq G$ be a non-empty subset.
Define $C_G(A) \overset{\text{def}}{=} \{g \in G \mid gag^{-1} = a \text{ for all } a \in A\}$, which is called the Centralizer,
and $N_G(A) \overset{\text{def}}{=} \{g \in G \mid gAg^{-1} = A\}$, which is called the Normalizer.

Prop. Let G be a group, and $A \subseteq G$ be a non-empty subset, and $H \leq G$,
1). $N_G(A)$ and $C_G(A)$ are subgroup of G.
2) The normalizer $N_G(H)$ is the largest subgroup such that H is normal.
3). A subgroup $H \leq G$ satisfies $H \leq C_G(H)$ if and only if H is abelian